

A Quaternionic Transport Interpretation of Holt Prime Constellations: Toward an Arithmetic Double Pendulum

Thomas Hoffbauer

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Abstract

We propose a transport-geometric interpretation of the nonconvex prime constellations studied by Fred B. Holt in the cycles of gaps $G(p^\#)$ of Eratosthenes sieve. Instead of viewing prime constellations merely as static admissible gap patterns, we interpret them as oriented transport paths on a discrete residue-state geometry derived from the modulo-12 prime classes. This yields a natural quaternionic holonomy structure, a parent-child coupling operator, and an arithmetic analogue of a coupled double pendulum. We formulate first structural lemmas and propose statistical tests for detecting non-isotropic long-range transport structure among admissible prime-gap configurations.

1 Introduction

Fred B. Holt recently analyzed admissible nonconvex prime constellations of type

$$(458, 3240), \quad (459, 3242),$$

within the cycles of gaps $G(p^\#)$ arising in Eratosthenes sieve [1]. These configurations evolve through primordial cycles via admissible residue transport.

The present note proposes a reinterpretation of these structures as

oriented transport systems

rather than merely admissible gap configurations.

The central idea is that the geometric structure is carried not by individual primes, but by transitions between admissible residue states.

2 EABC Transport Geometry

We define four distinguished residue sectors modulo 12:

$$E \equiv 1, \quad A \equiv 5, \quad B \equiv 7, \quad C \equiv 11 \pmod{12}.$$

A prime-gap constellation

$$s = (g_1, \dots, g_n)$$

induces a transport path

$$x_{k+1} = x_k + g_k \pmod{12}.$$

Thus every constellation defines a word

$$W(s) = (x_0, x_1, \dots, x_n)$$

in the discrete state space

$$\mathcal{S} = \{E, A, B, C\}.$$

3 Transition Matrices

Definition 1. For a constellation s , define its EABC transition matrix

$$M_s(a, b) = \#(a \rightarrow b),$$

where $a, b \in \mathcal{S}$.

This converts a prime-gap constellation into a discrete transport operator. The reversal symmetry of Holt constellations naturally yields:

$$M_{s^{\text{rev}}} = M_s^\top.$$

Hence the system possesses a discrete time-reversal symmetry.

4 Quaternionic Holonomy

We identify the EABC states with quaternionic units:

$$E \leftrightarrow 1, \quad A \leftrightarrow i, \quad B \leftrightarrow j, \quad C \leftrightarrow k.$$

Definition 2. The quaternionic holonomy of a constellation s is

$$Q(s) = \prod_{m=1}^n q(x_m),$$

where $q(x_m) \in \{1, i, j, k\}$.

Since quaternion multiplication is noncommutative,

$$ij = k, \quad jk = i, \quad ki = j,$$

the ordering of transitions carries genuine orientation information.

5 Parent-Child Coupling

Holt showed that every (459, 3242) constellation has the form

$$[2\hat{s}2],$$

while the (458, 3240) constellations arise by removing either the initial or terminal gap 2 [1].

Define:

$$P = [2\hat{s}2], \quad H = [2\hat{s}], \quad T = [\hat{s}2].$$

The removed boundary gap induces the transport

$$C \rightarrow E.$$

Lemma 1 (Boundary Coupling Operator). For all Holt parent constellations,

$$M(P) = M(H) + R_{CE},$$

where

$$R_{CE} = e_C e_E^\top$$

is a rank-1 boundary transport operator.

This operator acts as the arithmetic coupling joint of the system.

6 Arithmetic Double Pendulum

We define the arithmetic double pendulum as:

$$\mathcal{D} = (M, Q, R),$$

where:

- M is the EABC transport matrix,
- Q is the quaternionic holonomy,
- $R = R_{CE}$ is the boundary coupling operator.

The corresponding entanglement operators are:

$$\Delta_H = Q(P)^{-1}Q(H),$$

$$\Delta_T = Q(P)^{-1}Q(T).$$

Preliminary computations indicate:

$$\Delta_H = 1, \quad \Delta_T = k,$$

suggesting that head modes remain synchronized while tail modes acquire nontrivial chiral holonomy.

7 Chiral Transport Hypothesis

Hypothesis 1 (Chiral Double-Pendulum Hypothesis). *Among Holt prime constellations,*

$$P(\Delta_T \neq 1) > P(\Delta_H \neq 1).$$

Furthermore, the tail modes preferentially occupy nontrivial quaternionic axes

$$\{\pm i, \pm j, \pm k\}.$$

This would imply non-isotropic long-range transport structure within the primordial flow.

8 Discussion

The proposed framework suggests that:

prime constellations

may be interpreted as

topologically coupled transport resonators.

In this picture:

- admissible residues define local transport channels,
- quaternionic holonomies encode orientation,
- parent structures induce boundary couplings,
- reversal symmetry acts as discrete time reversal,
- and long prime-gap constellations behave as coupled arithmetic oscillators.

The essential open question is whether the induced quaternionic transport phases become asymptotically isotropic or exhibit statistically preferred axes.

9 Future Work

The next stage is numerical:

- compute holonomy distributions for all Holt constellations,
- analyze reversal-pair spectra,
- test isotropy of quaternionic axes,
- correlate asymptotic populations with transport spectra,
- and compare with genuine prime-gap data at large scales.

References

- [1] Fred B. Holt, *On Nonconvex Constellations Among Primes II: $(J, |s|) = (458, 3240)$* , arXiv:2605.19165 (2026).